World Health Organization

National Health Simulation Exercise Programme

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Glossary

- **7-1-7 Timeliness Metrics**: A framework to measure timeliness in detection, notification, and response to health emergencies within 7 days, 1 day, and 7 days, respectively.
- **Ad hoc Simulation Exercises (SimEx)**: Simulation exercises conducted sporadically without a scheduled plan, limiting their effectiveness in emergency preparedness.
- **Capability:** Demonstrated ability and practical competence of countries, systems, or organizations to mobilize and effectively utilize their capacities to achieve specific objectives related to prevention, preparedness, detection, response, recovery, and mitigation of health emergencies. Capabilities are reflected through observable, measurable, and validated performance during exercises, simulations, real events, or assessments, demonstrating operational effectiveness and readiness in managing health risks and emergencies, aligned with the core capacities identified by the International Health Regulations (2005).
- **Capacity Development**: Efforts to enhance the abilities, knowledge, and skills of individuals and institutions to perform emergency preparedness and response functions effectively.
- **Capacity:** Foundational resources and infrastructure required to effectively detect, assess, notify, and respond to health emergencies as outlined in the IHR. This includes human resources, health system infrastructures, laboratories, surveillance systems, logistical arrangements, legislation, financing, communication systems, and information management platforms. Capacities are the essential building blocks necessary for countries to achieve readiness, ensure early detection, rapid response, and effective management of public health risks and emergencies.
- **Confidence in Emergency Response**: The preparedness and self-assuredness of personnel to execute emergency protocols effectively under stress.
- **Emergency Operations Centre (EOC)**: A centralized facility for coordinating emergency management and response activities during health emergencies.
- **Fidelity**: The degree of realism in a simulation exercise, ranging from low fidelity (tabletop exercises) to high fidelity (full-scale field exercises).
- **Health Emergency Preparedness and Response (HEPR)**: A comprehensive approach to prevent, detect, and respond to health emergencies, emphasizing readiness and resilience.
- **Hot wash:** Immediate feedback or debriefing event involving the participants and the exercise management team.

- **IHR Monitoring and Evaluation Framework (IHR MEF)**: A formalized system under the IHR for assessing national capacities to manage public health emergencies.
- **Joint External Evaluation (JEE)**: A voluntary, collaborative, multisectoral process to assess country capacities to prevent, detect and rapidly respond to public health risks whether occurring naturally or due to deliberate or accidental events.
- **National Action Plan for Health Security (NAPHS)**: A country-owned, multi-year, planning process that can accelerate the implementation of IHR core capacities, and is based on a One Health for all-hazards, whole-of-government approach. It captures national priorities for health security, brings sectors together, identifies partners and allocates resources for health security capacity development.
- **National Health Simulation Exercise Programme (NHSEP)**: A structured, coordinated approach for delivering simulation exercises at a national level to enhance health emergency preparedness and response.
- **One Health Approach**: A collaborative, multisectoral approach to achieve optimal health outcomes by recognizing the interconnection between people, animals, plants, and their shared environment.
- **Preparedness and Resilience for Emerging Threats (PRET)**: A WHO initiative focused on improving pandemic preparedness using a group-based approach to health threats.
- **Simulation Exercise (SimEx)**: A practice activity designed to test, validate, and enhance emergency response plans and systems through real-world scenarios.
- **State Parties Annual Reporting Tool (SPAR)**: An annual self-assessment conducted by countries to evaluate their compliance with the International Health Regulations (2005).
- **Tabletop Exercise (TTX)**: A discussion-based simulation exercise involving decision-makers to test strategies and protocols in a controlled environment.
- **Universal health coverage (UHC)** means that all people have access to the full range of quality health services they need, when and where they need them, without financial hardship. It covers the full continuum of essential health services, from health promotion to prevention, treatment, rehabilitation and palliative care(1)

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List of abbreviations

DRR	Disaster Risk Reduction
HEPR	Health Emergency Preparedness, Response, and Resilience
IHR	International Health Regulations
IHR MEF	International Health Regulations Monitoring and Evaluation Framework
JEE	Joint External Evaluation
NAPHS	National Action Plan for Health Security
NDMA	National Disaster Management Agency
NHSEP	National Health Simulation Exercise Programme
NPHI	National Public Health Institute
PRET	Preparedness and Resilience for Emerging Threats
PHEOC	Public Health Emergency Operations Centre
RP	Responsible Person
SimEx	Simulation Exercise
SMART	Specific, Measurable, Achievable, Realistic, Time-bound
SPAR	State Parties Annual Reporting Tool
TTX	Tabletop (Simulation) Exercise
ToR	Terms of Reference

Executive Summary

Amid increasingly complex and unpredictable health emergencies, robust preparedness, readiness and response capabilities are paramount. Simulation exercises (SimEx) offer a powerful method to systematically test, validate, and enhance these capabilities. When implemented through a coordinated National Health Simulation Exercise Programme (NHSEP), SimEx strengthen a country's ability to manage health emergencies effectively, fostering readiness, interoperability, and resilience across the national health sector.

This guidance document provides a strategic framework, outlining the rationale, essential components, and practical steps for establishing, implementing, and sustaining an NHSEP. It champions a structured, integrated approach to SimEx, moving beyond sporadic initiatives to achieve more impactful and lasting preparedness, operational readiness, and measurable improvement in emergency response capabilities. The NHSEP aims to strengthen core capacities as defined by the International Health Regulations (IHR, 2005) and related global frameworks (e.g., the Health Emergency Preparedness, Response and Resilience (HEPR)(2) and the National Health Emergency Alert and Response (3) frameworks), thereby enhancing national readiness for all-hazards health emergency management.

Robust emergency management is achieved through three critical, sequential steps:

- **Step 1**: Establish and design the NHSEP: Building a strong foundation via dedicated leadership, comprehensive stakeholder engagement, and clearly defined strategic objectives.

- **Step 2:** Implement the NHSEP: Delivering a well-planned and systematically executed SimEx aligned with national priorities, identified risks, and existing health security plans.

- **Step 3**: Monitor and evaluate the NHSEP: Continuously tracking progress, assessing exercise effectiveness, and ensuring the integration of lessons learned into policy and operational practices.

The NHSEP process is sustained by high-level political commitment and continuous stakeholder engagement. It draws upon existing national preparedness goals, health policies, risk and capability assessments (such as the IHR States Parties Self-Assessment Annual Reporting Tool [SPAR](4) and Joint External Evaluation [JEE](5)), strategic preparedness and response plans, training needs, and insights from previous exercises. Ongoing monitoring and evaluation inform future implementation and strategic revisions, with secure, long-term commitment, including sustained funding, being crucial for programme continuity and efficacy.

This document guides those responsible for NHSEP development and implementation and encourages interoperability within the health sector and collaboration with other relevant sectors and agencies. Furthermore, it encourages systematic application of learnings and recommendations from SimEx.

The scope of this guidance covers SimEx applicable to all facets of health preparedness, readiness and response, from One Health initiatives and 7-1-7 timeliness metric assessments to testing warning systems for natural and human-induced hazards. It also covers activities such as hospital emergency surge capacity planning and Emergency Operations Centre (EOC) function testing.

While strategically comprehensive, this document does not detail methods for designing and delivering individual SimEx. For such practical guidance, readers should consult the WHO Simulation Exercise Manual (2017) (6). This guidance should be used in conjunction with other WHO resources, national SimEx materials, and existing emergency preparedness and readiness frameworks, ultimately contributing to improved global health security.

Chapter I: Background

1.1. Introduction

Simulation exercises (SimEx) are indispensable for strengthening health emergency preparedness and response. This document, "World Health Organization guidance on implementing a national health simulation exercise programme to strengthen emergency preparedness, readiness and response (NHSEP)", provides a comprehensive framework for countries to establish, implement, and sustain a robust NHSEP. The NHSEP aims to strengthen national capability to manage health emergencies through a structured series of SimEx. This guidance emphasizes an integrated approach, moving beyond ad hoc exercises, to foster a culture of continuous improvement and learning within the national health security system. It aligns with the International Health Regulations (IHR) (2005)(7), the Pandemic Agreement and other global health security frameworks like Health Emergency Preparedness, Response and Resilience (HEPR)(2) and the recently published National health emergency alert and response framework(3), aiming to build resilient health systems capable of effectively responding to diverse public health threats. The guidance details the rationale for an NHSEP, its potential benefits, and the key steps involved in its design, implementation, monitoring, and evaluation. It supports countries in developing a strategic, risk-informed, and sustainable programme that enhances interoperability, stakeholder collaboration, and the systematic application of lessons learned from SimEx. This guidance advocates for a cross-cutting or integrative function rather than a new vertical programme. The NHSEP should be integrated within an existing national authority, accompanied by a mandate on health emergency preparedness.

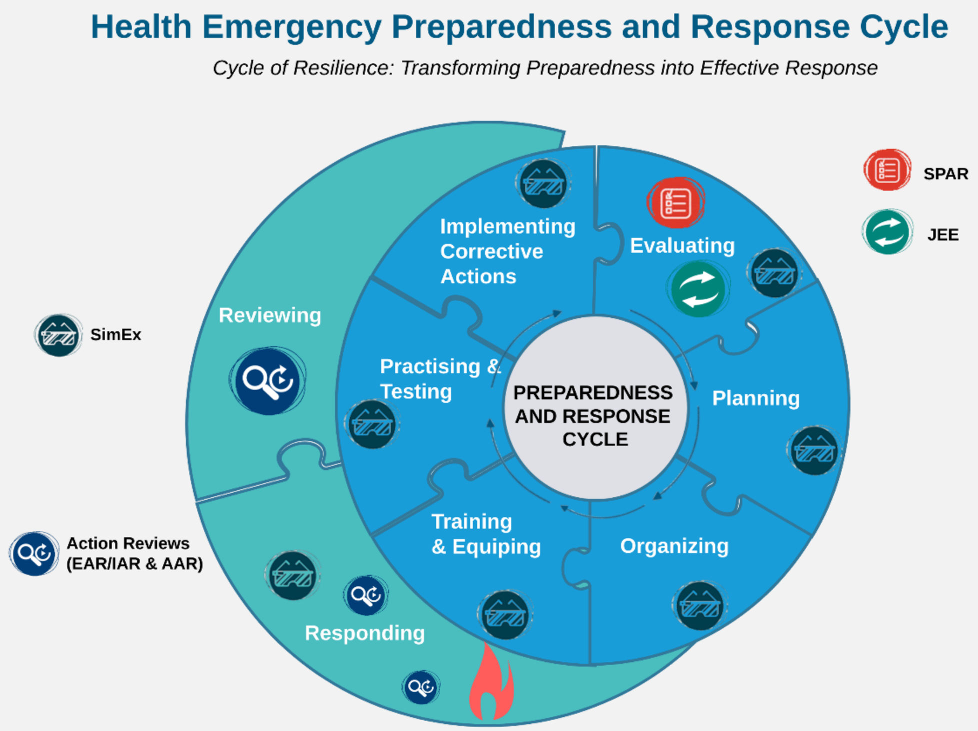

Figure 1. Health Emergency Preparedness and Response Cycle. Source: World Health Organization.

1.2. Who should use this guidance?

This guidance is primarily for government entities and partners responsible for establishing, delivering, and reporting on a NHSEP at national and subnational levels. This includes Ministries of Health (MoH) and National Public Health Institutes, but also authorities that often lead or coordinate national risk management such as National Disaster Management Agencies (NDMA), Ministries of Interior/Home Affairs, Civil Protection, Planning/Finance, and Prime Minister/President's Offices, as well as sectoral ministries (e.g., agriculture/animal health, environment, transport) and emergency operations centres (EOC/PHEOC). Target users include policymakers, emergency preparedness and response planners, public health officials, and SimEx coordinators. This includes policymakers, emergency preparedness and response planners, public health officials, and SimEx coordinators. Researchers and academics may also find this document useful as a reference. For countries with existing SimEx programmes, this guidance can serve as a resource to enrich their current efforts by providing a global perspective and standardized approach. Ultimately, it supports countries in identifying areas for corrective and preventive actions, strengthening best practices, and improving overall response performance to public health emergencies.

1.3. Why should countries conduct SimEx?

Simulation exercises (SimEx) are crucial for evaluating and enhancing a country's health emergency response capabilities. They provide a controlled environment to test plans and protocols, identify gaps, and improve the skills and confidence of personnel. Human resources are central to effective emergency management, and SimEx address the challenge of maintaining a skilled and confident workforce. While increasing workforce numbers through traditional training is often slow and costly, SimEx offer a practical way to enhance the competencies of existing personnel. Regular practice through SimEx helps staff retain skills, gain experience, and build confidence in executing emergency protocols under pressure. This is particularly important as experienced personnel retire and new threats emerge. The benefits of SimEx are well-documented, including improved planning, capacity building, and the maintenance of system, staff, and resource preparedness. By recreating the challenges of real emergencies, SimEx allow for the validation of plans and the development of critical competencies, ensuring that response personnel not only develop the requisite skills for navigating emergencies, but also confidence in their ability to act effectively during public health events.

The NHSEP also provides a structured mechanism for assessing capabilities, in line with WHO's emphasis on functional performance, by moving beyond capacity development to testing how effectively systems perform under real-world conditions. Importantly, NHSEPs can be embedded within or linked to broader national disaster risk management systems, depending on country context and existing governance structures. This flexibility enables countries to adopt models suited to their institutional maturity, whether under the MoH, NPHIs, or disaster management authorities.

While SimEx are valuable tools for workforce competency, each exercise also contributes to gap identification, system stress-testing, and policy validation. NHSEP consolidates these efforts into a cohesive programme to ensure national preparedness is progressively strengthened.

1.4. Why should countries have an NHSEP?

While individual SimEx are valuable, a NHSEP offers a more strategic and impactful approach to enhancing emergency preparedness. An NHSEP addresses common challenges associated with ad hoc SimEx, such as inconsistent alignment with national priorities, suboptimal implementation of recommendations, and difficulties in reporting and tracking progress. It also ensures that exercises are regularly conducted, address all core capacities outlined in the IHR (2005), and contribute to a continuous cycle of learning and improvement. Furthermore, an NHSEP facilitates accountability for implementing recommendations and allows for consistent reporting to WHO and other stakeholders, thereby maximizing the long-term benefits of SimEx for national health security. Lessons from SimEx should be used synergistically together with insights gained from action reviews to improve emergency management.

Countries may adopt different models based on their baseline capacities and existing structures. NHSEP can be initiated within national disaster management entities or other authorities and transition into MoH-led initiatives as capacities evolve. This flexible and contextual approach allows for joint ownership and avoids creating siloed systems.

1.5. Potential benefits of an NHSEP

A well-implemented NHSEP offers numerous benefits, contributing significantly to a country's health security. It facilitates strategic-level coordination and oversight of all SimEx, ensuring alignment with national preparedness priorities and the systematic allocation of adequate resources, including funding and staffing.

An NHSEP ensures that lessons identified through SimEx are effectively disseminated and incorporated into operational practices and policies, both within the health sector and across other critical sectors like agriculture, animal health, wildlife, environment and disaster management entities, thereby promoting cross-sectoral learning and preparedness. By maintaining a comprehensive overview of planned and completed exercises, NHSEP supports a progressive approach to SimEx complexity, ensuring regular testing of relevant risks and thorough validation of health system functional capabilities. It also systematically supports the practice and validation of interoperability of intersectoral plans and procedures, promoting a whole-of-society approach. Furthermore, an NHSEP enables a structured process for identifying and implementing corrective actions, leading to sustained enhancements in national capacities. Through standardized monitoring, evaluation, and reporting, it fosters greater accountability and performance measurement against national and international benchmarks, including the IHR. Ultimately, by systematically addressing diverse hazards, an NHSEP enhances resilience to multi-hazard and multisectoral threats, reducing vulnerability and improving response and recovery outcomes.

1.6. Risk-informed approach to SimEx

To maximize their effectiveness, SimEx within an NHSEP should adopt a risk-informed approach. This involves prioritizing risks for exercises based on comprehensive, evidence-based, all-hazards, multisectoral risk assessments. Such assessments enable countries to strategically target SimEx scenarios on relevant threats and their potential consequences, enhancing preparedness capabilities. When designing SimEx, planners must consider complex situations, including single hazards with cascading effects (e.g., floods leading to disease outbreaks) or simultaneous multi-hazard scenarios (e.g., earthquakes during conflict). Certain threats, like climate change-related emergencies, may have chronic impacts requiring sustained management. Key risk considerations for prioritization and scenario design include evaluating threats with severe local impacts (including those intensified by climate change), assessing seasonal risks and periods of concurrent emergencies, clearly identifying at-risk populations and geographical areas to tailor scenarios to local vulnerabilities, and leveraging risk assessment outcomes to evaluate existing capacities, governance and contingency financing structures, and coordination mechanisms. By embedding these risk-informed principles, countries can ensure targeted preparedness efforts, robust capability enhancement, and improved resilience across all sectors.

1.7. SimEx informed by the WHO Benchmarks for strengthening health emergency capacities

To maximize their strategic value, SimEx conducted within an NHSEP should also be guided by, and aligned with, the WHO Benchmarks for strengthening health emergency capacities(8). This i designing and implementing SimEx based on the capacities identified and prioritized through the WHO Benchmarks. By aligning SimEx objectives with the country's assessed capacity levels, whether foundational, developing, or advanced, exercise planners can ensure that scenarios are tailored to stress-test fundamental gaps, validate operational capabilities, and reinforce progressive benchmarks across its technical areas. This approach enables countries to focus on areas with low or intermediate capacity levels, such as workforce, emergency coordination, medical countermeasures, or public health and social measures (PHSM), while also integrating cross-cutting elements like gender equity, One Health, and community engagement.

WHO Benchmark-informed SimEx planning requires a systems-oriented perspective that tests not only individual technical functions but also the coherence of the IHR, the "five Cs" of the HEPR framework: collaborative surveillance, community protection, scalable care, access to countermeasures, and emergency coordination, and the National health emergency alert and response framework. Exercises should simulate realistic scenarios that test capacities and capabilities like intersectoral coordination and multi-level governance, expose bottlenecks in surge capacity or supply chain operations, and examine linkages across the national and subnational response architectures. Additionally, WHO Benchmark-informed SimEx can be used to validate the effectiveness of prior capacity-building interventions and identify where further investments or policy revisions are needed. By grounding exercises in benchmark actions and progression pathways, countries can ensure that simulation activities are not isolated events but integral components of a continuous learning and system-strengthening cycle, and ultimately fostering more resilient and responsive health emergency systems.

Chapter II: Framework and Structural Elements of the NHSEP

2.1. NHSEP design, implementation and evaluation steps

Developing and implementing an NHSEP involves three interconnected steps:

- **Step 1**: Establish and design an NHSEP, which includes establishing the programme, designing its overall structure, and developing a national SimEx schedule.
- **Step 2**: Implement the NHSEP, which focuses on planning and delivering SimEx according to the established schedule.
- **Step 3**: Monitor and evaluate the NHSEP, which involves tracking programme implementation and evaluating its effectiveness in improving IHR capacities.

These steps are complementary, and individual steps may be pursued concurrently with others. For instance, existing SimEx activities can continue while the broader NHSEP is being formally established. The overarching aim is to cultivate a culture of exercising capabilities against known risks, thereby ensuring improved readiness and adequate resource allocation.

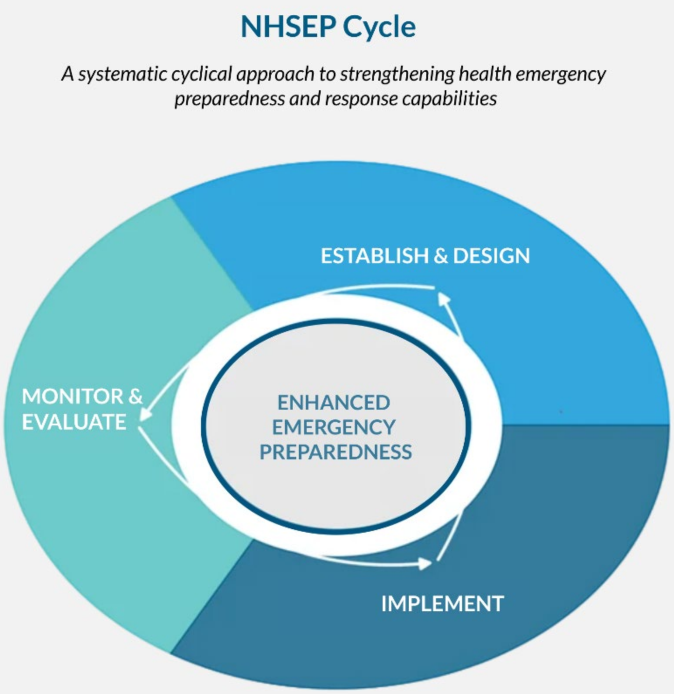

Figure 2. NHSEP Cycle.

2.2. Key functions and activities of an NHSEP

While each national programme will be unique, an NHSEP should generally encompass several key functions and activities. These include developing a country-contextualized framework, policy document, and operational plan for the SimEx programme. It also involves establishing a SimEx coordination group and identifying a simulation exercises leader to coordinate the programme. Securing a sustainable funding stream and necessary resources, including logistics, is crucial. Furthermore, identifying or developing a pool of trained SimEx experts or emergency response personnel who can be trained to serve as SimEx experts is essential for the programme's success. For countries with limited resources, adapting the programme, sharing lessons, and pooling resources across jurisdictions can help overcome challenges related to dedicated personnel and governance structures.

2.3. Key components of an NHSEP

Several key components are required to develop a comprehensive NHSEP. These include:

- A strategy document outlining the programme's vision and goals, and an operational plan detailing its implementation.
- Terms of Reference for both the SimEx coordination group that define roles and responsibilities.
- Clearly defined programme aims and objectives, aligned with available emergency preparedness and response plans, risk and capacity assessment and review results (including the SPAR and JEE outcomes), and a gender-inclusive and responsive approach, are fundamental.
- A detailed SimEx schedule, budget, and relevant supporting plans are also critical components.
- Consideration of relevant national and subnational legal instruments (e.g., health or emergency management policies) to embed SimEx requirements into statutes, aligning with IHR and The Pandemic Agreement expectations for sustained capacity and legal support for their implementation.

Chapter III: Step 1. Establish and Design an NHSEP

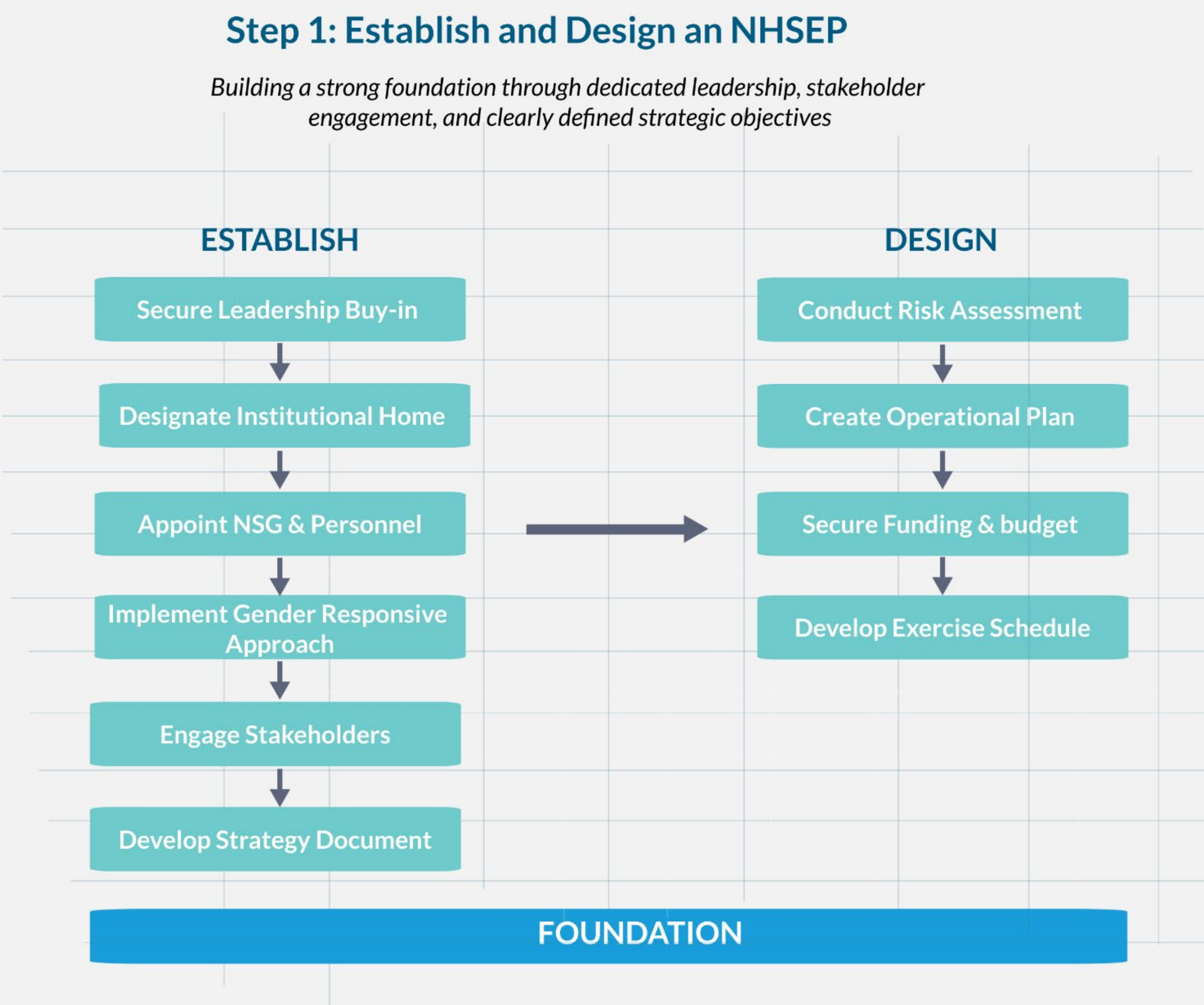

Figure 3. Establishing and Designing an NHSEP.

3.1. Establish an NHSEP

Establishing a NHSEP requires several key administrative and strategic actions before the detailed design phase. This foundational stage involves securing political and strategic engagement, designating an institutional home, appointing key personnel, ensuring inclusivity, engaging stakeholders, drafting a strategy document, estimating funding, and obtaining official endorsement. An NHSEP should be developed into the core of the national emergency preparedness strategy. It can be placed within existing structures, such as emergency response departments, in line with directions from the MoH and aligned with country plans, the IHR, and response networks.

3.1.1. Introduce the NHSEP concept and secure leadership buy-in

Initiating an NHSEP requires strong political and strategic commitment. Senior leadership within the Ministry of Health (MoH) or an equivalent authority should act as champions for the programme. Their support is critical for mobilizing the necessary time, staff, and resources, and for encouraging participation in SimEx across all levels of the health system.

When introducing the NHSEP, it is important to clearly articulate:

- what SimEx are and their role in the preparedness cycle,
- their benefits compared to other training methods,
- the rationale for a national programme and its alignment with national and global priorities,
- how a risk-informed approach will be adopted,
- expected resource requirements and key stakeholders, and
- governance, evaluation, and improvement mechanisms, including long-term impacts and sustainability.

To ensure maximum impact, the NHSEP should be integrated into national health emergency preparedness and response policy frameworks and anchored in a clear legal or executive basis. This may take the form of legislation, a ministerial decree, or an executive order, which would formalize objectives, mandate regular simulation exercises, and establish governance structures. Establishing such a foundation at the outset strengthens political commitment and creates the conditions for institutionalization and sustainability.

3.1.2. Designate an institute

Once leadership buy-in is secured and a legal or policy framework is initiated, the next step is to designate an institutional home for the NHSEP. The programme should be embedded within a national authority that has a mandate over health emergency preparedness, response coordination, and inter-sectoral collaboration. This ensures legitimacy, alignment with national strategies, and long-term sustainability.

Possible institutional placements include:

- the MoH (e.g., Department of Emergency Preparedness or Health Security) as the primary lead;
- a Public Health Emergency Operations Centre (PHEOC) for operational aspects;
- a National Disaster Management Agency (NDMA) to support inter-sectoral coordination; or
- National Public Health Institutes (NPHIs) or designated IHR Authorities.

Institutional designation should be explicitly linked to the legal or policy framework, ensuring that the chosen entity has a formally recognized mandate to lead the NHSEP and convene relevant stakeholders. This alignment prevents overlaps, clarifies authority, and provides a statutory anchor for continuity, accountability, and resource allocation. Once the framework and NHSEP strategy document (see Step 3.1.6) are finalized, formal endorsement by senior MoH officials and other relevant authorities (e.g., NDMA, Ministry of Finance, Prime Minister's Office) will reinforce both political commitment and statutory mandate.

3.1.3. Appoint relevant persons

Dedicated leadership is vital. A Simulation Exercises Coordination Group (SimEx Coordination Group), led by a Simulation Exercises Leader, should guide each step of the NHSEP. For consistency, the term "SimEx Coordination Group" will be used throughout this guidance. The SimEx coordination group should be small, manageable, and engage external experts as needed. Ideally, the work should be led by its key tasks include seeking funding, ensuring resource availability, and overseeing the programme. The appointment process should follow the mandate set in the legal or executive instrument, ensuring that roles and responsibilities are formally recognized within institutional structures. A simulation exercises leader, typically a senior manager within the MoH with experience in emergency preparedness and knowledge of the health sector (and potentially links to an NDMA), should lead the NHSEP. This leader could be a National SimEx Coordinator, focal point within MoH, or a specific new role depending on the country context. This individual should provide direction, advocate for the programme, and engage with stakeholders to ensure its success and the implementation of lessons learned.

Table 1 lists the minimum members recommended to form a SimEx coordination group, ensuring balanced representation across key health and emergency institutions.

Table 1: Suggested minimum composition of the SimEx coordination group.

- Simulation Exercises Leader
- Administrative support
- MoH emergency preparedness manager
- Senior hospital manager
- Senior public health manager
- Senior pre-hospital care/ambulance manager
- Representative of the National Disaster Management Agency (NDMA)
- Representative from a non-health sector to support wider uptake and understanding (optional)

The core functions and responsibilities that should guide the work of the SimEx coordination group are summarized in Table 2.

Table 2: Suggested Terms of Reference for the SimEx coordination group.

- Identify ambassadors for the programme
- Identify agencies that need to be made aware of and/or participate in the programme
- Identify focal points at subnational levels and others who may be a main entry point for the introduction of the programme
- Develop the NHESP strategy document based on the WHO Guidance on Implementing an NHSEP in conjunction with a panel of specialists identified by the SimEx coordination group
- Develop the NHSEP operational plan
- Develop the NHSEP schedule
- Seek out internal and external funding/donor opportunities
- Ensure sufficient personnel are trained to design, develop and deliver simulation exercises
- Appoint a person who will have overall responsibility for developing and managing the programme
- Oversee the three steps of the NHSEP

Table 3 highlights the leadership duties expected of a Simulation Exercises Leader to ensure effective management and advocacy of the NHSEP.

Table 3. Suggested Terms of Reference for the Simulation Exercises Leader.

▪ Chair the SimEx coordination group ▪ Appoint members of the SimEx coordination group ▪ Lead the development of the NHSEP and delegate authority if and when required ▪ Develop key components of the NHSEP ▪ Develop, organize and lead the planning workshops ▪ Execute the operational plan ▪ Disseminate the SimEx results to relevant authorities ▪ Coordinate funding activities ▪ Liaise with stakeholders and encourage advocacy ▪ Have knowledge of the health sector – public health, emergency response and emergency preparedness

While leadership roles such as the Simulation Exercises Leader and the SimEx Coordination Group are essential for launching and guiding the NHSEP, long-term sustainability depends on integrating simulation activities into the regular responsibilities of relevant institutions. Embedding SimEx planning, delivery, and follow-up into existing job descriptions, protocols, and institutional performance systems ensures continuity beyond individual assignments. This helps make simulation exercises a routine part of national emergency preparedness, rather than a separate or temporary function.

3.1.4. Ensure an inclusive and gender-responsive approach

Social norms and vulnerabilities can lead to different experiences for various social groups, including by gender, during health emergencies. The NHSEP design must incorporate considerations at both strategic and operational levels based on gender as well as considering other vulnerable groups including older persons, people with disabilities, marginalized groups, and those with chronic illnesses. This includes ensuring diverse representation in the National Steering Group (NSG), developing SimEx scenarios that reflect gendered impacts (e.g., access to care for pregnant women), ensuring diverse participant selection in exercises, disaggregating data by sex for analysis, advocating for inclusive communication strategies, considering gender-specific logistical needs (e.g., facilities), and explicitly addressing gender issues in debriefing sessions. WHO's Gender Equity and Human Rights in Emergencies Toolkit can provide further resources.

Suggested gender and inclusion indicators for NHSEP:

- Percentage of participants from underrepresented groups (by gender, age, disability, minority status).
- Number of exercise scenarios explicitly addressing gender-specific vulnerabilities.
- Proportion of post-exercise reports with sex-disaggregated data.
- Number of recommendations from SimEx that address gender or inclusion gaps.

3.1.5. Engage key partners and stakeholders

Mapping and engaging stakeholders and key partners across sectors is essential for collaboration and leveraging diverse expertise and resources. Health organizations and the broader health sector should actively participate in SimEx with relevant stakeholders, civil society and community leaders, (e.g., agriculture, food safety, transport, NDMA, private providers, minority groups, disability advocates, voluntary sector, faith-based organizations) as emergency responses are rarely isolated. This multisectoral involvement ensures exercises reflect real-world complexities.

Developing links with external partners enhances information sharing and understanding of different agency response mechanisms. Stakeholder mapping tools from WHO can be useful in this process.

3.1.6. Draft the NHSEP strategy document

The strategy document is a contextualized guiding framework tailored to the specific country. It articulates the necessity of SimEx and defines the strategic purpose, aims, scope, and objectives of the NHSEP, providing a clear roadmap. The strategy should also build upon the draft or updated legal and policy framework developed under Step 3.1.2, ensuring that its objectives and governance arrangements are anchored in statutory or executive provisions. It should outline strategic priorities aligned with national preparedness goals, detail programme direction (including staffing and resource allocation), and incorporate findings from national risk assessments and existing emergency plans. It should also consider existing SimEx programmes for synergy.

This guidance should be informed by, and aligned with, national and multisectoral risk assessments, strategic preparedness plans, and established frameworks, including those related to health security, disaster risk reduction, and whole-of-society approaches such as One Health and civil–sector collaboration. Comprehensive risk assessment processes that evaluate the likelihood, severity, and exposure of hazards are essential for prioritizing simulation exercise (SimEx) scenarios, identifying high-risk subnational areas, and structuring multisectoral engagement. The findings from these assessments should guide the alignment of the NHSEP with national risk profiles, ensuring that exercises effectively test priority hazards, critical functional capacities, and coordination mechanisms.

A model outline of what an NHSEP strategy document might contain is provided in Table 4 to guide national planners.

Table 4. Suggested outline of an NHSEP strategy document.

Title: NHSEP Strategy document for Country X

- Executive summary
- Background/Introduction/Overview
- Purpose, aims, scope and objectives
- Alignment with national goals and strategies, including existing risk and preparedness capacity and readiness capability assessments, emergency preparedness and response plans; outcomes of action reviews (such as early action reviews [EAR], intra action reviews *[*IAR*]* and, after action reviews *[*AAR*]*) (9)
- Sector-wide and stakeholder analysis/SWOT analysis
- Operational principles
- Strategic directions – including proposed timeline
- Financial proposal
- Evaluation methods
- Annexes (e.g., Responsible Person and NSG membership, Terms of Reference, Monitoring and Evaluation tools)

3.2. Design the NHSEP operational plan

Following the establishment of the NHSEP, the next phase is to develop a detailed operational plan. This plan translates the strategic objectives from the strategy document into concrete, actionable steps, specifying activities, timelines, responsibilities, and resources required for effective programme implementation. It serves as a practical guide for organizing and delivering SimEx, ensuring alignment with the NHSEP's overall goals and the country's emergency preparedness priorities. The operational plan includes a schedule of exercises, personnel requirements, budgetary allocations, and mechanisms for monitoring and evaluation, fostering consistency, accountability, and measurable progress.

3.2.1. Convene an NHSEP planning workshop

Planning workshops are instrumental in developing and mapping out the NHSEP operational plan. Attendees should include the SimEx coordination group, representatives from other relevant government departments and agencies, senior health emergency preparedness personnel, and operational staff with relevant local and/or regional knowledge. Workshop objectives include developing yearly operational plans at the start of a 2-5 year NHSEP cycle and reviewing the implementation of these plans during the cycle.

Table 5 proposes discussion topics that can help structure and enrich an NHSEP planning workshop.

Table 5. Suggested discussion topics for the planning workshop.

▪ What risks should be covered by the SimEx schedule? ▪ Which currently existing emergency plans will be tested and/or validated? ▪ Who is willing to be an ambassador for the NHSEP within their organization? ▪ Which agencies need to be made aware of and/or participate in the programmes? ▪ How can engagement at both the national and local levels be ensured? ▪ How will the simulation exercises be staffed and resourced? ▪ What types of SimEx will be conducted? ▪ Who would be the target participants for various levels of simulation exercises? ▪ What would a schedule of simulation exercises look like at both local and national levels? ▪ How should the NHSEP be monitored and evaluated (e.g., what does success look like and how is it measured?)

3.2.2. Formulate specific aim, objectives, scope, and timeline

The operational plan must have a clearly stated aim (what the NHSEP hopes to achieve) and SMART objectives (i.e., specific, measurable, achievable, realistic, time-bound) that incorporate a gender-inclusive approach. The scope should align with the national emergency health sector's size and capacity, including subnational considerations, IHR capacities, geographical areas, and the specific emergency phase to be assessed. All IHR capacities should be assessed at least once during the 2–5 year NHSEP cycle. The timeline for an NHSEP cycle is typically 2-5 years, allowing for effective risk identification, exercise planning, and information sharing, depending on resource availability and political will.

Objectives should incorporate a gender-responsive approach to health emergency preparedness. An example of how aims and SMART objectives could be framed within an NHSEP operational plan is illustrated in Table 6.

Table 6. Example of an NHSEP operational plan aim and corresponding objectives.

Aim. To develop, implement, and review an NHSEP to improve health system preparedness and response capacity that covers the emergency health sector in *[your country]*.
Objectives ▪ Develop a national health SimEx strategy document covering a period of *[number]* years. ▪ Ensure that adequate resources, including funding and staffing, are allocated to SimEx over *[number]* years. ▪ Implement the health SimEx schedule as stipulated in the SimEx strategy document. ▪ Continuously evaluate (and adjust where necessary) the implementation of the SimEx schedule. ▪ Ensure that lessons identified and learnt from health SimEx are disseminated and incorporated into policy and operational practice within the health sector and extended to other sectors as required.

3.2.3. Outline personnel requirements

Assess whether necessary capacity (e.g, staff, resources, time) is available. For less-experienced personnel, starting with simpler SimEx (e.g., tabletop exercises) is recommended to build skills and confidence for more complex scenarios. A pool of experienced SimEx practitioners is central to success. As complexity increases, additional personnel such as exercise managers, control staff, facilitators, and evaluators are needed. Training for personnel involves two components: 1) training for planning and conducting SimEx (i.e., building a pool of experts through courses like the WHO Simulation Exercise training); and 2) training for staff participating in SimEx (i.e., ensuring emergency health-care staff are trained for their roles in an all-hazards approach).

3.2.4. Prepare a budget

The operational plan must include a budget estimating costs for implementing programme activities for its duration including for logistics support. Sample budget templates are available in the WHO SimEx Manual and accompanying toolkit (10).

Financing should cover not only SimEx delivery but also governance, institutional support, coordination, and learning processes. NHSEP outputs can support advocacy for resource allocation within NAPHS and broader health security budgets. Simulation exercises can further help prioritize NAPHS actions, identify costed gaps, and inform investment cases for donors and ministries of finance.

The budget should therefore address both direct SimEx costs (logistics, materials, venue, etc.) and broader requirements such as human resources for coordination, governance structures, training, monitoring and evaluation, and integration activities. It should also consider overhead, sustainability planning, and alignment with NAPHS financing pathways.

3.2.5. Draft schedule with individual exercises and their delivery

The operational plan includes a detailed schedule of individual SimEx (discussion-based or operational). This schedule outlines the timeline, scope (i.e., risks covered), types (i.e., modality, complexity, fidelity), and frequency of exercises. When planning SimEx, ensure that they align with health preparedness priorities, address emerging and/or heightened risks, test plan amendments, validate new capacities, and incorporate lessons from previous simulations. A risk management approach should identify priority areas. The schedule should also consider the time required to develop SimEx, resources, and participant availability, and incorporate local/regional exercises.

The scope of SimEx should be risk-based. The modality and complexity of these exercises can range from simple drills to complex full-scale exercises. Depending on time and resource availability, SimEx fidelity can be low (e.g., tabletop exercises) or high (e.g., field simulations), with both being beneficial. High-fidelity exercises, for example, are more complex but can lead to more effective assessments and adoption of best practices. SimEx frequency may range from quarterly to biannual, with low-fidelity exercises ideally being conducted more often than high-fidelity ones (i.e., at least one high-fidelity exercise per 2-5 years programme). Sample SimEx schedules focusing on plans, frequency or risk, and JEE and/or SPAR outputs are provided in the original WHO guidance and may be adapted to country contexts as needed.

Table 7 sets out a suggested outline for an operational plan, showing how a country might translate its NHSEP strategy into practice.

Table 7. Suggested outline of an NHSEP operational plan for Country X.

Title: NHSEP operational plan for Country X, year xxxx to year xxxx • Background, purpose and scope • Alignment with NHSEP strategy • Objectives, strategies and their expected results • Individual exercises and their schedule of delivery • Assigned tasks and timelines • Budget and resources • Monitoring and evaluation • Assumptions and risks • Ongoing feedback, guidance, and advice

Key design questions that planners should consider when preparing individual simulation exercises are shown in Table 8.

Table 8. Questions for individual exercise planning.

• What are the aim/purpose, objectives, and scope of the exercise? • Who are the intended participants of the exercise? • Who will staff the exercise? • What scenario will the exercise be based on? • What will we name the simulation exercise? (NB: It is best to avoid political or religious references) • Is there a planning team familiar with the selected scenario, or does the planning team need to consult with a relevant subject matter expert? • Who are the stakeholders in the exercise? • Who will prepare the post-exercise report and who will it be distributed to? • Who is the person responsible for ensuring that identified lessons are implemented?

Table 9 provides a checklist to help planners build a coherent and well-prioritized SimEx schedule.

Table 9. Checklist for developing a SimEx schedule.

Activity to be completed	Checkbox
• Align SimEx with health preparedness priorities, addressing new or heightened risks from emerging or existing hazards • Exercises should test amendments to emergency plans, validate newly developed capacities, and incorporate lessons from previous simulations to address identified gaps • Collaboration with other sectors may also inform exercise priorities, particularly for testing cross-sectoral coordination • This approach ensures that exercises are relevant, targeted, and contribute meaningfully to enhancing health system readiness and resilience	
• Undertake a risk management approach to identify SimEx priority areas that need immediate attention	
• Classify which risks and priority areas need to be addressed or validated, and determine how frequently they should be assessed (e.g., six months, annually or every two years)	
• Align preparedness priorities with the findings of risk assessments	

• Preparedness priorities are overarching goals established for a specific timeframe, such as five years, and are informed by risk assessments that identify critical threats and vulnerabilities • More complex operational exercises, conducted every three years, provide opportunities to test the entire emergency response system under realistic conditions • By linking preparedness priorities to the risks highlighted in assessments, this approach creates a continuous improvement cycle where each activity builds on the insights of previous efforts, ensuring a more resilient and responsive health system	
• Undertake a planned roll out of the NHSEP, considering the length of time it takes to develop a SimEx, the resources involved, and the need for relevant participants to attend	
• Incorporate local and regional exercises as part of the NHSEP	
• Identify an individual or department responsible for each exercise	
• Verify that relevant health sector areas and plans are included in the NHSEP, such as for laboratories, hospitals, points of entry, mass casualties, outbreak response, business continuity, natural hazards, and emergency medical teams (EMT)	
• Verify that the NHSEP exercise schedule lists progressively increasing complex exercises	
• Allocate an appropriate name for each SimEx or group of SimEx	
• Verify that sufficient and appropriate resources have been allocated to conduct SimEx. For example, these may include a trained pool of staff, information technology equipment availability in EOCs, funding for each SimEx, and resources for follow-up actions	

Chapter IV: Step 2. Implement the NHSEP

Once the NHSEP structure is established, the governance and legislative frameworks identified, and the exercise schedule prepared, the implementation phase begins. This step focuses on the ongoing support and actions needed to execute and maintain the NHSEP, with designated health agencies taking responsibility for delivering the individual SimEx.

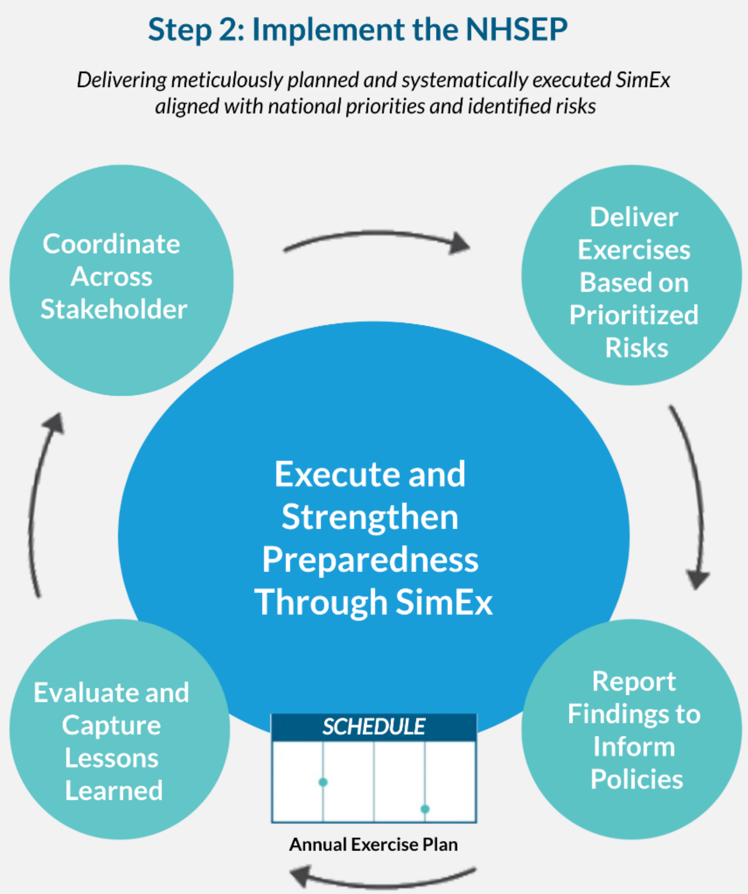

Figure 4. NHSEP Implementation.

4.1. Convene coordination meetings

Regular coordination meetings of the SimEx coordination group are essential for overseeing the implementation of the operational plan. These meetings, ideally held 3-4 times a year and tailored to country context), ensure sustained engagement from senior stakeholders and high-level decision-makers, maintaining focus on the SimEx schedule.

Ongoing issues to discuss include funding, delivery schedule adherence, availability of trained staff, implementation activities, and arranging workshops or conferences to share lessons from SimEx. Routine maintenance of emergency response plans and risk assessments can also impact the NHSEP schedule; monitoring these changes will identify areas for updates. Inputs from SimEx debrief sessions, including specific attention to gender-related issues, should inform the redesign of subsequent exercises. The NSG will review and adjust the operational plan quarterly and yearly as needed.

4.2. Deliver individual SimEx according to schedule

The SimEx coordination group is responsible for implementing the SimEx schedule. Exercises should be executed as planned to maintain consistency and achieve intended objectives. Exceptions should only occur under specific circumstances, such as an active emergency in the exercise locality or new information from capability assessments necessitating adjustments. In such cases, the schedule should be reviewed and modified, not cancelled, to uphold NHSEP goals and maintain capacity-building momentum.

4.3. Conduct post-exercise surveys and evaluation

While indicators and reporting strategies are typically designed during SimEx planning, implementing agencies must ensure these pre-defined indicators are actively applied and evaluated during the exercise to objectively measure objective achievement. This involves collecting real-time data, observations, and participant feedback (i.e., as outlined in the WHO SimEx Manual and Toolkits). This information allows for a thorough assessment of exercise effectiveness and provides actionable insights for future improvements.

4.4. Prepare post-exercise reports for national planning

Following each SimEx, a comprehensive post-exercise report must be prepared. This report should detail the exercise's objectives, activities, outcomes, identified strengths, areas for improvement, and SMART recommendations for corrective actions. These reports are crucial for informing national planning processes, guiding policy adjustments, and ensuring that lessons learned are integrated into broader emergency preparedness and response strategies. The reports should be disseminated to all relevant stakeholders to foster shared learning and accountability.

4.5. Prepare a SimEx minimum reporting protocol as part of the IHR

SimEx are a voluntary component of the IHR MEF(11). For a SimEx to be recognized within this framework, a minimum set of information must be reported to WHO. Countries should use a standardized template to report on their NHSEP activities annually. This reporting ensures that SimEx contributions to health emergency preparedness and capacity building are consistently documented and acknowledged, supporting national accountability and global health security efforts.

To support standardized reporting to WHO, an example of a minimum NHSEP reporting template is included in Table 10.

Table 10. Example of a SimEx minimum reporting template.

Country name: xxx
NHSEP cycle: 2026–2028
Reporting period: From .... to ....
Reporting date:
Reporting officer: Name, Title

Minimum information required for all recommendations									
Date of the assessment (required)	Public health event (i.e., scenario) exercised (required)	Recommendations (i.e., operational and capacity development activities) (required)	Technical area (required)	Indicator (required)	Activity type	Start date (required)	End date (required)	Responsible Person or team	Activity feasibility

Chapter V: Step 3. Monitor and evaluate the NHSEP

The final step of the National Health Simulation Exercise Programme (NHSEP) focuses on the critical processes of monitoring and evaluating the implementation of the operational plan and its recommendations. This involves assessing the NHSEP's effectiveness, reviewing its relevance and content, and disseminating SimEx findings through comprehensive post-exercise reporting. Systematic monitoring and evaluation ensure objectives are met and help identify gaps or duplications in the SimEx schedule. Periodic reviews are essential to maintain alignment with evolving risk assessments, national policies, and global health security frameworks, ensuring the programme remains relevant and responsive. Monitoring and reporting also contribute to broader initiatives like the JEE, SPAR, and donor accountability, strengthening the NHSEP's integration into national and international health security strategies. Monitoring should occur regularly (i.e., quarterly, biannually, or annually), with reports shared among relevant agencies and stakeholders to foster transparency, collaboration, and continuous improvement.

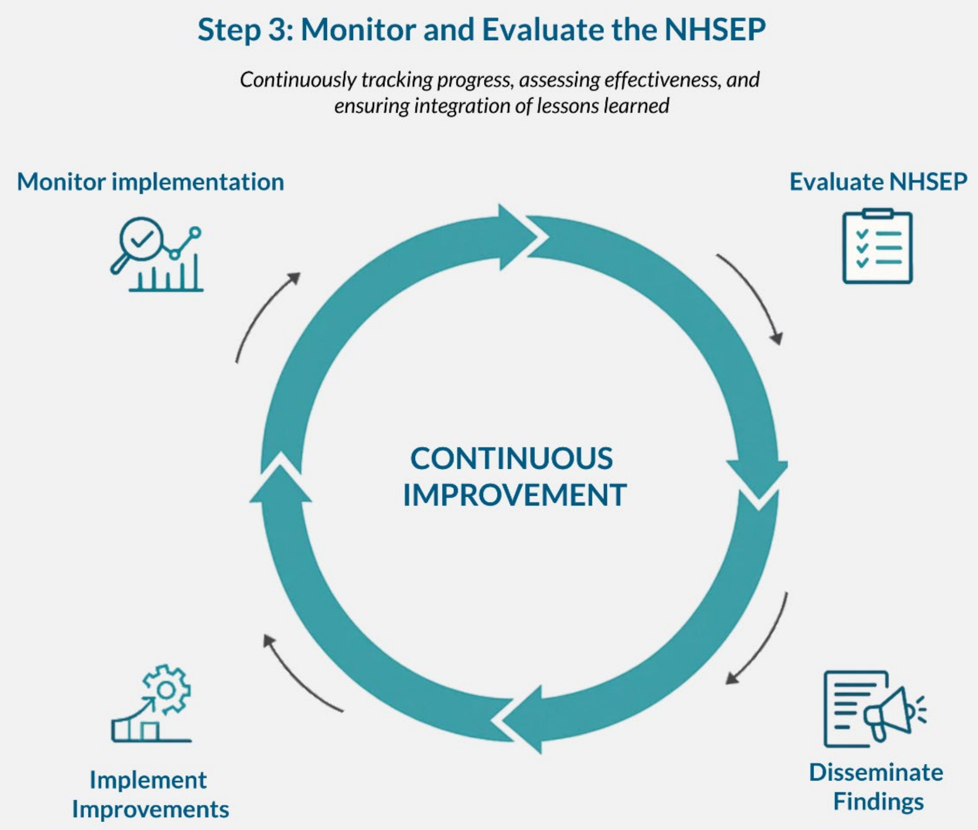

Figure 5. NHSEP Monitoring and Evaluation.

5.1. Monitor the implementation of the NHSEP operational plan

Monitoring the NHSEP operational plan involves several key aspects to ensure its effective execution and impact.

5.1.1. Adherence to the SimEx schedule

The SimEx schedule must be monitored to check agency compliance. If SimEx are not delivered as planned, mitigating factors should be identified to determine if scheduling improvements or additional resources are needed. This should occur regularly (e.g., monthly) and after each SimEx. A tracking system for individual SimEx delivery should be part of the NHSEP

Table 11 offers a simple monitoring checklist for tracking the implementation and reporting of simulation exercises.

Table 11. Example of an implementation and reporting checklist for monitoring SimEx.

SimEx name/ID	Proposed date of delivery	Actual date of delivery	SimEx report completed	Action plan Responsible Person	Date actions completed
Exercise Red Hat	*Day, month, year*	*Day, month, year*	*Day, month, year*	*Person Z*	*Day, month, year*

5.1.2. Linking SimEx monitoring to IHR/SPAR/JEE

Progress against the SimEx schedule and recommendations should be mapped to relevant International Health Regulations (IHR, 2005) indicators, including SPAR and JEE scores. This alignment will:

- Support advocacy for resources by showing direct contributions of SimEx to strengthening national health security capacities.

- Demonstrate compliance with, and progress towards, global health security benchmarks.
- Provide evidence for inclusion in national reports to WHO, donors, and other partners.

This mapping should be updated regularly and reviewed during NHSEP monitoring meetings, ensuring SimEx outputs feed into broader preparedness assessments.

5.1.3. Quality of exercises delivered

The NSG must monitor the quality of SimEx to ensure they meet a sufficient standard for developing and maintaining emergency preparedness. Quality monitoring covers exercise delivery, planning activities, and participant feedback and evaluation. Regular contact with regional and/or local health managers regarding SimEx helps identify problems early for remedial action. Feedback forms (refer to the sample in the WHO SimEx Manual) can be used. The designated individual should brief authorities on how SimEx address needs identified in risk assessments and emergency plans, making recommendations if certain areas are overlooked.

5.1.4. Inclusion (uptake) of SimEx recommendations into plans and policy development

Implementing recommendations from the SimEx Action Plan is critical. Successful integration is more likely when recommendations align with health sector priorities, collaborators are engaged early, and co-learning is promoted. The NHSEP should offer practical, actionable recommendations and support policy adjustments. Regular review of emergency plans is essential. Barriers to implementation (e.g., overly optimistic expectations, inadequately collaborative policymaking) should be proactively identified and addressed by the NSG. Examples of such barriers, along with potential solutions, are outlined in Table 12. A system for recording lessons and/or recommendations and tracking their completion is needed. Lessons should be shared widely and SimEx reports should be distributed, including to relevant stakeholders who were unable to participate.

Recommendations from SimEx should be tracked for their integration into national policies, operational guidelines, and budget submissions. Establishing a recommendation-to-action pipeline (e.g., uptake tracker, policy review mechanism) supports tangible improvements in systems and capabilities.

Table 12. Barriers to implementing SimEx recommendations.

Barriers to implementing SimEx recommendations include: • A lack of funding to implement the required changes • A lack of resources to implement the required changes • A lack of buy-in from senior staff to agree to the changes • Limited organizational knowledge about how to implement the changes • A lack of staff with relevant expertise in technical areas to inform the changes the existence of cultural issues preventing the implementation of recommendations.

5.1.5. Implementation of SimEx recommendations articulated in policy documents

The NHSEP should track not only the uptake of SimEx recommendations into policy documents but also their actual implementation. Including recommendations in national or sub-national plans like the National Health Sector Strategic Plan and/or processes like NAPHS (via the e-NAPHS platform)(12) can garner high-level support for capacity-building and learning, and mobilise resources.

The NHSEP should also evaluate IHR capacities after recommendations are implemented. A combined tracker for recommendation uptake and implementation, aligning with NAPHS and NHSEPs, may be an effective way to track the implementation of SimEx recommendations.

Table 13 presents a tracker template that countries can use to monitor how recommendations from SimEx are adopted and implemented.

Table 13. SimEx recommendations uptake and implementation tracker.

Minimum information required for all recommendations									
Date of the assessment (required)	Public health event (i.e., scenario) exercised (required)	Recommendations (i.e., operational and capacity development activities) (required)	Technical area (required)	Indicator (required)	Activity type	Start date (required)	End date (required)	Responsible Person/team	Activity feasibility (Low-Medium-High)

5.1.6. Administrative performance

Monitoring administrative performance assesses the efficiency of tasks supporting smooth programme running. This includes adherence to the exercise schedule (i.e., logistics, meeting scheduling), delivering quality exercises (i.e., equipment, expert selection, partner involvement), and documentation of recommendations (i.e., report quality, communication, stakeholder collaboration).

5.2. Evaluate the NHSEP

Evaluation is crucial for the NHSEP's evolution and continued relevance.

5.2.1. Evaluation purpose

An NHSEP is dynamic and should adapt to evolving health emergency preparedness needs and risk assessments. Evaluation is typically conducted to determine a programme's efficiency, effectiveness, relevance, impact, sustainability, and replicability. Regularly reviewing exercise effectiveness in building capabilities is vital. The national SimEx schedule needs assessment to ensure it remains fit for purpose and considers changes in risks, policies, and resources.

5.2.2. Determine the evaluation interval

NHSEP evaluation should occur at planned intervals, such as midterm and end-of-cycle (e.g., after 2–3 years for a 5-year programme, and at the end of the 5 years). The timing should align with national planning cycles (e.g., NHSEP and NAPHS review). Ad hoc evaluations may be needed after major public health events or following significant changes in national priorities or risk profiles.

5.2.3. Plan and conduct the evaluation

This involves several sub-steps:

- **Evaluate feasibility**: Assess if an evaluation is practical, considering data availability, resources, and stakeholder buy-in.
- **Develop an evaluation team**: Form a team with diverse expertise (SimEx design, M&E, health systems, relevant technical areas), including internal and external members for objectivity.
- **Implement the evaluation**: Define scope and questions, select methods (document reviews, interviews, surveys, case studies), collect and analyze data, and synthesize findings.

5.2.4. Disseminate evaluation findings

Share findings widely with stakeholders (e.g., MoH, NSG, partners, donors) through reports, presentations, and workshops. Ensure that findings are accessible and understandable, highlighting successes, challenges, and actionable recommendations.

5.2.5. Follow up on evaluation recommendations

Develop an action plan to address recommendations, assign responsibilities, set timelines, and monitor implementation. Integrate lessons into the next NHSEP planning cycle to ensure continuous improvement.

The Responsible Person or their team should ensure recommendations meet the criteria in Table 13 to facilitate their implementation.

Criteria that make evaluation recommendations actionable and practical are outlined in Table 14.

Table 14. Criteria for evaluation recommendations.

Criteria for evaluation recommendations include the following. ▪ Recommendations should follow logically from conclusions, lessons learned and any emerging good practices ▪ Recommendations should be numbered and concisely written as stand-alone concepts, without unexplained acronyms, that can be responded to individually by line management ▪ Specify who is called upon to act ▪ Specify the action needed to remedy the situation ▪ Distinguish the level of priority (i.e., high, medium, or low) ▪ Specify the time frame for follow-up (i.e., short-term, medium term, long-term or not applicable) ▪ Acknowledge whether the proposed recommendations require resource mobilisation

Note: These best practices were adapted from the International Labour Organization's guidance on managing recommendations from independent project evaluations(13).

Table 15 illustrates a management response system evaluation tool that can be adapted to track follow-up actions on recommendations.

Table 15. Management Response System Evaluation.

Management response	**Action plan**	**Progress made**	**Comments**	**Information source**	**Date of line management response**
Completed	*(free text)*	*(free text)*	*(free text)*	*(free text)*	*(free text)*
Partially completed, action not yet taken					
No action planned					

Note: These best practices were adapted from the International Labour Organization's guidance on managing recommendations from independent project evaluations (13).

5.3. Prepare an NHSEP strategy document for the next planning cycle

At the end of an NHSEP cycle, a new strategy document for the next cycle should be prepared. This process should ideally begin 6-12 months before the current cycle ends. The NSG and the Responsible Person are responsible completing this task. The new strategy should be informed by the evaluation findings of the current NHSEP, updated risk assessments, national health priorities, IHR MEF outputs (i.e., SPAR, JEE, AR), lessons from actual emergencies and SimEx, and stakeholder feedback. The preparation involves reviewing past performance, consulting stakeholders, drafting the new strategy (revisiting aims, objectives, scope, priorities), and obtaining endorsement. Anticipated outcomes include a revised, evidence-informed NHSEP strategy, improved alignment with national/global health security agendas, enhanced stakeholder engagement, and a clear roadmap for continued capacity building.

5.4. Establish a centralized digital repository for SimEx documentation and recommendations

This repository should be maintained by the NHSEP coordination group secretariat or designated national authority and include all exercise reports, action plans, evaluation findings, and implementation status updates. The repository should be accessible to relevant stakeholders and integrated with national health information systems to support institutional memory, transparency, and continuous learning.

Conclusion

The COVID-19 pandemic and other recent global health emergencies have underscored the importance of systematic, continuous, and rigorous preparedness through the use of simulation exercises (SimEx). By establishing and maintaining a National Health Simulation Exercise Programme (NHSEP), countries can proactively test, validate, and enhance their capacities and capabilities for managing public health emergencies effectively. This guidance has highlighted how structured and strategically coordinated SimEx significantly contribute to national, regional, and global health security through strengthening core capacities defined under the IHR and HEPR.

An NHSEP is not merely a mechanism for individual exercises; rather, it is a strategic platform designed to embed sustainable learning and continuous improvement into national preparedness and response frameworks. As demonstrated throughout this guidance, effective SimEx align closely with comprehensive national assessments, strategic preparedness goals, and global frameworks, enabling countries to strengthen interoperability, resilience, and whole-of-society collaboration. By systematically incorporating corrective actions, countries ensure sustained progress and meaningful improvements in emergency management practices.

The NHSEP promotes a systematic risk-informed approach, prioritizing exercises based on comprehensive risk assessments, including scenarios exacerbated by climate change, cascading hazards, and complex multi-hazard situations. Moreover, SimEx complement action reviews, providing an essential mechanism for preparedness even in the absence of actual emergency events. This synergy ensures that national health systems remain robust, prepared, and adaptable to evolving threats.

Furthermore, establishing an NHSEP positions countries to actively participate in and benefit from international cooperation and peer-to-peer learning opportunities, reinforcing global health security and collective resilience. Countries are encouraged to continuously evaluate their NHSEP implementation, ensuring its relevance, effectiveness, sustainability, and responsiveness to emerging threats and priorities.

By adopting this WHO guidance on implementing an NHSEP, countries will not only enhance national preparedness but also contribute significantly to a safer, healthier, and more resilient world.

Looking forward, NHSEPs should evolve to address emerging challenges such as climate change-related health emergencies, antimicrobial resistance, AI-driven health misinformation, and complex humanitarian crises. Integrating these evolving risks into future SimEx design will ensure that national systems remain agile, forward-looking, and resilient.

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Annex:

I. Methodology for the Development of the Guidance

This guidance document was developed through a systematic and consultative process to ensure its relevance, applicability, and technical soundness. The methodology was designed to capture a wide range of perspectives and experiences from global, regional, and country levels, ensuring that the final product is both evidence-informed and field-tested.

1.1. Multi-level and Multi-stakeholder Consultation

The development of this guidance was a collaborative effort involving various stakeholders. The process included extensive consultations with:

- World Health Organization (WHO): Input was sought from all three levels of the organization, WHO's Headquarters, Regional Offices, and Country Offices, to ensure the guidance aligns with strategic priorities and is practical for implementation in diverse settings.
- Member States: Several Member States were engaged to provide feedback based on their unique country contexts, needs, and experiences with simulation exercises and health emergency preparedness.
- Technical Advisory Group on Simulation Exercises: The WHO Technical Advisory Group on Simulation Exercises, comprised of leading international experts, provided critical technical review and expert input throughout the development process.

1.2. Global Consultative Meeting

A key component of the development process was the Simulation Exercises and Action Reviews Consultation Workshop, held in Berlin, Germany, from 25–28 April 2023. The meeting was hosted at the WHO Hub for Pandemic and Epidemic Intelligence and brought together representatives from WHO Headquarters, Regional and Country Offices, Member States, partner agencies, academic institutions, and international public health organizations.

The consultation included two main workstreams:

- **Simulation Exercises**: Focusing on the review and refinement of the draft Guidance for the Implementation of a National Health Simulation Exercise Programme;

- **Action Reviews**: Concentrating on advancing methodologies for early, intra-, and after-action reviews (EAR/IAR/AARs) and exploring synergies across global partners.

The objectives of the meeting were to:

- Review and update the draft guidance on the Implementation of a National Health Simulation Exercise Programme;
- Discuss the implementation process for national simulation exercise programmes across Member States;
- Define next steps for the global rollout of action review methodologies, including the forthcoming COVID-19 AAR guidance;
- Explore the concept and next steps for the Nuggets of Knowledge (NoK) Platform as a collaborative mechanism for experiential learning in health emergencies.

Participants represented a diverse cross-section of expertise in health emergency preparedness, epidemiology, public health security, and response management, including colleagues from WHO's six regional offices, partner agencies such as Africa CDC, ECDC, UKHSA, US CDC, and Resolve to Save Lives, as well as technical experts and consultants supporting WHO's work on simulation exercises and action reviews.

Members of the WHO Technical Advisory Group on Simulation Exercises also participated in this global consultation, providing technical expertise and strategic input to further strengthen the guidance and ensure its alignment with global best practices.

The workshop provided valuable inputs on refining the draft guidance on the Implementation of a National Health Simulation Exercise Programme, and aligning it with broader efforts to strengthen the Health Emergency Preparedness, Response, and Resilience (HEPR) architecture.

1.3. Iterative Development and Review

The guidance was developed in an iterative manner, with multiple rounds of drafting and review. Feedback and input from stakeholders across WHO levels and partner institutions were systematically collected, analyzed, and incorporated into successive versions of the document. This iterative approach ensured that the final guidance is comprehensive, user-friendly, and reflects the collective knowledge and experience of the global health security community.

II. Members of the Technical Advisory Group on Simulation Exercises (July 2021 to June 2023)

Name	Title/Position	Organization/Affiliation	Role
Dr George Karagiannis	Engineering Leadership Group Director	Resilience First, United Kingdom	Chair
Ms Lucia Mullen	Senior Analyst, Associate	Johns Hopkins Center for Health Security	Co-chair
Mr Anthony Craig	UN World Food Programme Representative	Global Protection Cluster - Mine Action Area of Responsibility	Member
Dr Noel Lee J. Miranda	Regional Technical Consultant	Public Health Emergency Management, Asia Pacific	Member
Dr Mustafa Özyavuz Kerem	Co-director	Marmara University Hospital, Türkiye	Member
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Professor Limor Samimian-Darash	Academic Officer	School of Public Policy and Governance, Hebrew University of Jerusalem	Member
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